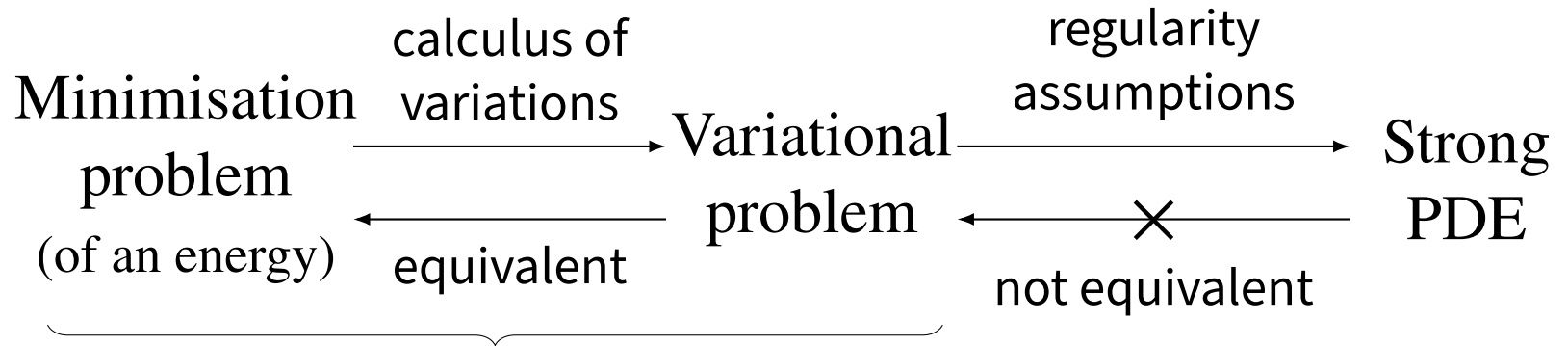


Chapter 1 — Mathematical models

From energy minimisation to PDEs

MTH3340 · Numerical Methods for PDEs

1 Introduction

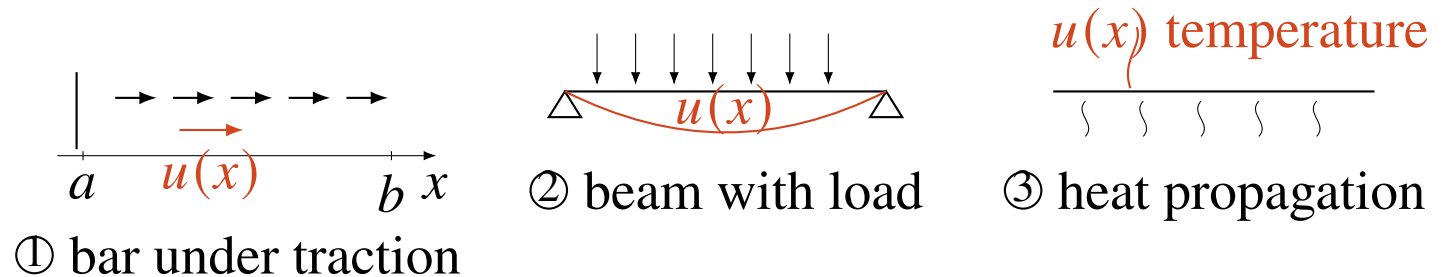


$\exists!$ of solutions via functional analysis

Why is this important?

- The finite element method is based on the variational problem.
- The variational problem admits solutions when the strong PDE does not make sense (lack of regularity).

2 Minimisation problem



$u(x)$ is the unknown, a displacement or a temperature. E.g., for ①,

$$E(u) = \frac{1}{2} \int_a^b k(x) u'(x)^2 dx$$

is the internal (elastic) energy, and $k(x)$ is the Young's modulus.

Data:

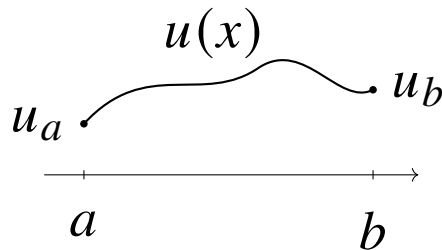
- boundary conditions $u(a) = u_a, \quad u(b) = u_b,$
- forcing term $f : [a, b] \rightarrow \mathbb{R}$ (load).

Minimum energy principle

Total energy

$$J(u) \doteq \underbrace{\int_a^b \frac{1}{2} k(x) u'(x)^2 dx}_{\text{internal energy}} - \underbrace{\int_a^b f(x) u(x) dx}_{\text{external work}}$$

* In which space do we want to minimise?



- We need to satisfy the boundary conditions.
- We need to evaluate $u'(x)$, so we require $u \in C^1([a, b])$.

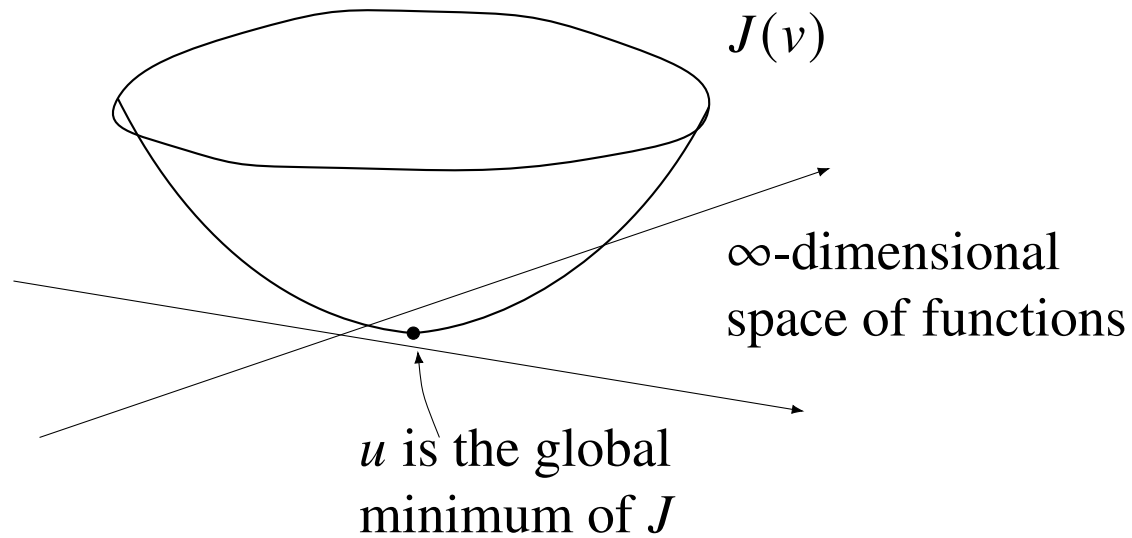
Minimisation problem:

$$u \doteq \arg \min_{\substack{v \in C^1([a, b]) \\ v(a)=u_a, v(b)=u_b}} J(v)$$

∞ many functions in this set!

A minimisation problem in ∞ dimensions.

The idea, in a picture



3 Variational formulation

Definition (Vector space)

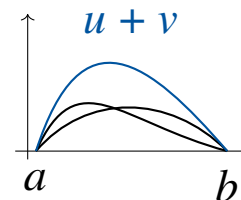
A vector space V of real-valued functions on a domain $\Omega \subset \mathbb{R}^d$ (d the space dimension) is such that, for any $u, v \in V$ and $\alpha \in \mathbb{R}$, $u + v \in V$ and $\alpha u \in V$, with

$$(u + v)(x) \doteq u(x) + v(x), \quad (\alpha \cdot u)(x) \doteq \alpha u(x), \quad \forall x \in \Omega.$$

Example: the space $C_0^1([a, b])$ (the subscript 0 means that the functions vanish at the boundary) is a vector space,

$$u, v \in C_0^1([a, b]) \longrightarrow u + v \in C_0^1([a, b]), \quad \alpha u \in C_0^1([a, b]), \quad \alpha \in \mathbb{R}.$$

Same structure as vectors in $\mathbb{R}^n$ (linear algebra).



Q: Is the set of functions $u \in C^1([a, b])$ with values $u(a) = u_a$, $u(b) = u_b$ a vector space?

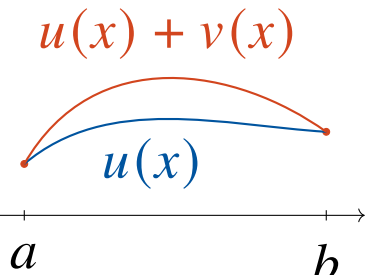
Calculus of variations

Assume there exists a global minimum $u(x)$ of J (the minimisation problem).

Compute variations of J at $u(x)$ in the direction $v(x) \in C_0^1([a, b])$.

Note: $u(x)$ and $v(x)$ are functions,

$$\begin{cases} u(x) \in C^1([a, b]), \\ u(a) = u_a, \quad u(b) = u_b, \end{cases}$$

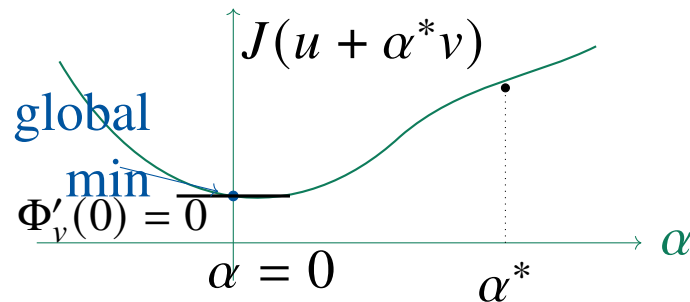
$$v(x) \in C_0^1([a, b]).$$


Then, $u(x) + \alpha v(x) \in C^1([a, b])$ and it satisfies the boundary conditions. It is an *admissible* function in our minimisation problem.

We can define the function of one real variable

$$\Phi_v(\alpha) \doteq J(u + \alpha v).$$

Stationarity at the minimum



By definition, Φ_v has a global minimum at $\alpha = 0$. Assuming Φ_v is differentiable,

$$\Phi'_v(0) = 0.$$

I.e., since u is the global minimum, any perturbation can only increase J .

Now, let us compute this derivative using the definition of derivative of a function,

$$g'(x) = \lim_{\varepsilon \rightarrow 0} \frac{g(x + \varepsilon) - g(x)}{\varepsilon}.$$

Computing the variation

$$\Phi'_v(0) = \lim_{\varepsilon \rightarrow 0} \frac{\Phi_v(\varepsilon) - \Phi_v(0)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{J(u + \varepsilon v) - J(u)}{\varepsilon}.$$

Using our J in the physical model above,

$$J(u + \varepsilon v) = \frac{1}{2} \int_a^b k \underbrace{\left[(u + \varepsilon v)' \right]^2}_{(u')^2 + 2\varepsilon u'v' + \varepsilon^2 (v')^2} dx - \int_a^b f(u + \varepsilon v) dx,$$

$$J(u + \varepsilon v) - J(u) = \varepsilon \int_a^b k u' v' dx + \frac{\varepsilon^2}{2} \int_a^b k (v')^2 dx - \varepsilon \int_a^b f v dx.$$

Dividing by ε and letting $\varepsilon \rightarrow 0$,

$$\Phi'_v(0) = \int_a^b k u' v' dx - \int_a^b f v dx = 0,$$

Variational problem / weak form of the PDE

$$\int_a^b k u' v' dx = \int_a^b f v dx \quad \forall v \in C_0^1([a, b]).$$

The converse also holds

We have proved that a global minimiser solves the variational problem. **Is a solution of the variational problem a minimiser?**

We already have the full expansion; Φ_v is a parabola in α ,

$$\Phi_v(\alpha) = J(u) + \alpha \left(\int_a^b k u' v' - \int_a^b f v \right) + \frac{\alpha^2}{2} \int_a^b k (v')^2.$$

$\Phi_v''(\alpha) = \int_a^b k (v')^2 > 0$ for $v \neq 0$; each Φ_v is a **convex parabola**. If u solves the variational problem, the linear term vanishes,

$$J(u + \alpha v) - J(u) = \frac{\alpha^2}{2} \int_a^b k (v')^2 > 0 \quad \forall v \neq 0, \alpha \neq 0,$$

so u is the *unique* global minimiser.

Equivalence

minimisation problem $\iff$ variational problem
(we can move freely between the two)

4 Differential equation

Lemma (Fundamental lemma of calculus of variations)

If $g \in C^0([a, b])$ satisfies $\int_a^b g(x)v(x) dx = 0$ for all $v \in C_0^\infty((a, b))$ (smooth functions vanishing near a and b), then $g \equiv 0$.

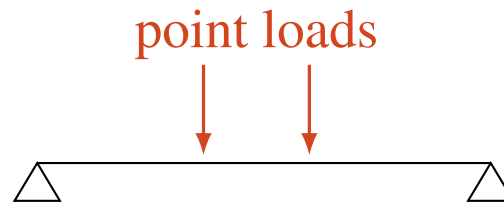
Assumption: the global minimum $u \in C^2([a, b])$ (not only C^1), $k \in C^1([a, b])$, $f \in C^0([a, b])$. Then, by integration by parts, for all $v \in C_0^1([a, b])$,

$$\begin{aligned}
 0 &= \underbrace{\int_a^b k u' v' dx - \int_a^b f v dx}_{\text{variational form}} \quad (\text{boundary term vanishes, } v = 0 \text{ at } a, b) \\
 &= - \int_a^b (k u')' v dx - \int_a^b f v dx = \int_a^b \left(\underbrace{-(k u')' - f}_{g \text{ in the lemma}} \right) v dx.
 \end{aligned}$$

By the lemma, $\boxed{-(k u')' = f \quad \forall x \in \Omega}$ strong PDE.

Weak vs. strong

There will be problems that admit a solution of the variational problem (or the minimisation problem) but not of the strong PDE.



E.g., for a point load, $f \notin C^0([a, b])$, and the strong problem does not make sense. The variational problem still does; we will make this precise in the next lectures.